

**CASE STUDY REPORT**

**on**

**Enhancing Personalized Skincare  
Recommendations Through AI and E-  
Commerce Integration: A Comprehensive  
Analysis**

Submitted in partial fulfilment for the course

**AECC27**

**Design Thinking**

**Bachelor of Engineering**

**in**

**CSE (Artificial Intelligence and Machine Learning)**

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## **CERTIFICATE**

This is to certify that **Mukul Prasad (USN: 1MS24CI076) and Ayesha Quresh (USN: 1MS24CI028)** have successfully completed the Case Study titled **“Enhancing Personalized Skincare Recommendations Through AI and E-Commerce Integration: A Comprehensive Analysis”** in partial fulfilment of the requirements of the course **AECC27 Design Thinking** offered by Ramaiah Institute of Technology, Bengaluru in partial fulfillment for the award of Bachelor of Engineering in **CSE (Artificial Intelligence and Machine Learning)** of the Visvesvaraya Technological University, Belagavi during the year 2024-25.

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### **Marks Allotment:**

| <b>Total Marks</b> | <b>Marks Obtained</b> |
|--------------------|-----------------------|
|                    |                       |

**Signature of Faculty with Date**

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# 1. Abstract

The rapid growth of the skincare industry has highlighted the need for personalized product recommendations to cater to diverse skin types and individual concerns. Traditional e-commerce platforms often fall short by offering generalized suggestions, resulting in subpar user satisfaction. This case study explores how integrating Artificial Intelligence (AI) with e-commerce can address these challenges. The analysis investigates the architecture, key features, and business benefits of AI-driven skincare recommendation platforms. Supported by recent literature, the study concludes that AI-enhanced e-commerce not only improves user engagement and product suitability but also boosts brand loyalty and conversion rates in the beauty sector.

# 2. Introduction

The skincare industry has evolved dramatically, with rising consumer demand for products tailored to specific needs. However, with countless product choices and lack of expert guidance, many users are left confused. Conventional e-commerce models provide limited personalization. Leveraging AI can bridge this gap by offering real-time skin analysis and customized product suggestions based on user input and behavioral patterns. This case study explores the efficacy of AI-driven recommendation systems integrated with e-commerce platforms and evaluates their role in reshaping the beauty industry.

Objectives of the Study:

- Understand AI recommendation mechanisms in skincare.
- Identify challenges in building personalized systems.
- Analyze integration of AI with e-commerce interfaces.
- Predict future trends and improvements.

Scope of the Study:

The study focuses on AI-powered skincare platforms with web-based interfaces aimed at diverse user demographics.

# 3. Literature Review

| Study / Author(s)  | Methodology                  | Key Contribution                                            | Tools / Tech Used     | Identified Gaps                                  | How This Study Adds Value                                |
|--------------------|------------------------------|-------------------------------------------------------------|-----------------------|--------------------------------------------------|----------------------------------------------------------|
| Han et al., 2018   | Deep Learning (Experimental) | Classified human scalp conditions using neural nets         | CNN, Image Processing | Focused only on scalp, limited dataset           | Applies concept to broader facial skincare needs         |
| Lee et al., 2020   | Machine Learning             | Developed a skin-care recommendation system using user data | Sensors, ML           | No real-time skin analysis, static data          | Explores real-time, camera-based assessment              |
| Forbes, 2021       | Industry Insight             | Discussed AI's role in beauty personalization               | AI, Beauty E-commerce | General insights, no technical implementation    | Provides technical architecture and use cases            |
| McKinsey, 2022     | Market Survey                | Showcased personalization's business value in beauty        | Analytics             | Lacks discussion of ethical and technical issues | Addresses ethical deployment and design integration      |
| Shopify Blog, 2023 | Trend Analysis               | Reported beauty e-commerce strategies                       | E-commerce Platforms  | No AI or personalization focus                   | Focuses on AI-e-commerce integration for personalization |
| Kumar et al., 2022 | Deep Learning (Experimental) | Built model for skin disease classification                 | CNN                   | Focused on clinical diagnosis, not e-commerce    | Bridges diagnosis with real-time product matching        |

|                     |                   |                                                       |                   |                                                   |                                                         |
|---------------------|-------------------|-------------------------------------------------------|-------------------|---------------------------------------------------|---------------------------------------------------------|
| Jalal et al., 2021  | Literature Review | Outlined AI in dermatology & clinical use             | Imaging, ML       | Regulatory and privacy barriers                   | Applies AI within consumer-safe and private framework   |
| Wang et al., 2021   | Systematic Review | Compared recommendation algorithms                    | RecSys Algorithms | No user personalization or evaluation standards   | Introduces adaptive recommendations with feedback loop  |
| Esteva et al., 2017 | Deep Learning     | Dermatologist-level accuracy in skin cancer detection | Deep Neural Nets  | Requires medical validation, not consumer-focused | Repurposes AI for consumer-level recommendation systems |

## 4. Research Methodology

This case study follows a **qualitative research approach**, focusing on understanding the real-world application and impact of AI-based skincare recommendation systems in e-commerce settings.

### □ Type of Research

#### 🔍 Qualitative Exploratory Study

The goal was to explore emerging technologies and user experiences surrounding AI in skincare, rather than to test a hypothesis or build a quantitative model.

### 📝 Data Collection Methods

#### 1. Secondary Research:

- Review of academic journals, whitepapers, and industry case studies related to AI in dermatology, e-commerce, and personalization systems.

## 2. Primary Research (Informal Interviews):

- Conducted **interviews with 5–6 students and peers** (both male and female, aged 18–22) from the same institution.
- Questions included:
  - “What do you find confusing while shopping for skincare online?”
  - “Would you trust AI-based skincare recommendations?”
  - “Do you face issues matching products to your skin type?”
- The responses helped uncover practical user pain points and expectations from such platforms.

## □ Tools Used

- **Literature Databases** (IEEE Xplore, Google Scholar) – for gathering peer-reviewed articles.
- **Web Development Environments**
  - *Frontend*: React, Next.js
  - *Backend*: Node.js
  - *ORM*: Prisma
- **Comparative Analysis Frameworks** – to benchmark features of current AI skincare tools.

## 📌 Procedure

1. **Literature Review:**

Reviewed existing research and case studies to understand how AI is being applied in skincare and e-commerce.

2. **Interviews & Observation:**

Conducted short discussions with students to gather firsthand insights into their skincare shopping behavior and concerns.

3. **Platform Analysis:**

Examined popular AI skincare apps and platforms to assess features, flow, and personalization depth.

4. **Gap Identification:**

Compared existing solutions against user feedback to find areas of improvement.

5. **Synthesis & Proposal:**

Combined findings to suggest a scalable AI-powered solution with real-time skin analysis and product recommendation.

## **5. Results / Findings**

**As students working on this case study, we focused on understanding how AI could make skincare shopping more helpful and personalized, especially for young users like ourselves. Our findings are based on the interviews we conducted, analysis of existing platforms, and our own testing and observations.**

### **Key Findings from Peer Interviews**



| <b>Insight Area</b>                | <b>What We Learned</b>                                                                                   |
|------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Confusion in product choice</b> | <b>Most peers didn't know what suits their skin and relied on trial &amp; error.</b>                     |
| <b>Overload of options</b>         | <b>Students often feel overwhelmed by too many choices on e-commerce websites.</b>                       |
| <b>Lack of expert input</b>        | <b>Many expressed they wanted dermatologist-like advice without visiting one.</b>                        |
| <b>Trust in AI</b>                 | <b>Surprisingly, most said they would trust an AI tool <i>if</i> it felt reliable and user-friendly.</b> |
| <b>Awareness of skin type</b>      | <b>Many weren't confident about their own skin type or conditions.</b>                                   |

### **Observations from Platform Analysis**

**We studied 4–5 popular skincare and beauty apps (like SkinVision, YouCam, Sephora's AI tools) and noticed:**

- **Some used AI to scan skin through the camera, but didn't explain the result clearly.**
- **Product recommendations felt generic on some platforms, not tailored.**
- **Many apps were designed for global users, but lacked support for Indian skin tones or local products.**

### **Outcomes & Patterns Identified**

- **AI shows strong potential in improving the skincare buying experience, especially with real-time analysis and conversational interfaces.**
- **Users prefer a simple, relatable, and trustworthy interface over something overly clinical or robotic.**
- **Personalization matters more than just a fancy UI — users want product suggestions that actually match their skin and budget.**
- **Our project also revealed that students are open to new tech if it helps them shop smarter.**

### **⚠ Limitations We Faced**

- **We worked with a small sample size (just a few peers), so results aren't fully generalizable.**
- **Time and resource constraints meant we couldn't develop a working app — just a prototype and concept.**
- **We didn't test real-time camera analysis due to lack of access to trained ML models.**

### **✅ Final Thought**

**Our study shows that even with limited tools, it's possible to identify real user problems and design a helpful AI-based solution. If developed further, this concept could genuinely improve the skincare experience for young, digital-first users like us.**

## **6. Discussion / Analysis**

### **□ What the Results Tell Us**

The responses from our interviews and analysis helped us understand how first-time skincare buyers, especially students, engage with e-commerce platforms. Many students face confusion, rely on generic product suggestions, or skip buying altogether due to lack of clear guidance. These results confirmed our assumption that there's a real gap in how current platforms serve users who aren't skincare experts.

The fact that many of our peers expressed interest in using an AI-based solution shows that digital-native users are open to tech-based assistance, as long as it feels trustworthy and relatable.

### Relating Our Findings to Previous Research

| Prior Research      | Their Focus                                         | What We Noticed Differently                          |
|---------------------|-----------------------------------------------------|------------------------------------------------------|
| Lee et al., 2020    | Personalization via sensors and ML                  | Users want <i>easy-to-use</i> , not tech-heavy tools |
| McKinsey, 2022      | AI boosts engagement and loyalty in beauty brands   | Our peers also value guidance over loyalty           |
| Wang et al., 2021   | Rec systems lacked personalization metrics          | We observed students want context-aware advice       |
| Esteva et al., 2017 | AI classifying skin diseases with clinical accuracy | Our focus is on everyday skin needs, not medical     |

Our study added to this by bringing a student-centric view and focusing on simple AI-powered features that would help real users, even those who don't know their skin type or ingredients.

### Unexpected Insights

- Even though AI seems complex, users trust it more when it's explained in a friendly, casual tone — not technical jargon.
- Many students said they've never been asked about their skincare struggles— this shows that even basic features like a "chat-based quiz" would add huge value.
- A few people mentioned they've used influencer advice or Instagram reels for skincare — this wasn't part of our initial scope but became an interesting theme for future exploration.

### ⚠ Challenges and Limitations

| Challenge                 | Impact                                                              |
|---------------------------|---------------------------------------------------------------------|
| Small interview group     | Only ~6 students, not enough for broad generalization               |
| No real AI implementation | We explored concepts, not full machine learning models              |
| Privacy limitations       | Couldn't fully test camera input due to ethical and tech boundaries |
| Platform study limits     | Public apps don't reveal much about backend AI logic                |

Despite these, we were able to develop a concept that's grounded in real user needs and supported by trends in industry research.

## 7. Conclusion

This case study explored how Artificial Intelligence (AI) can enhance personalized skincare recommendations within e-commerce platforms — especially from the perspective of students and first-time skincare buyers.

### □ Key Findings Recap

- Most students struggle to choose skincare products due to lack of guidance or understanding of their skin type.
- Existing platforms offer personalization, but it often feels generic, overly commercial, or not suited for Indian users.
- Peer interviews revealed that students are **willing to trust AI tools**, especially when presented in a friendly, helpful, and clear manner.
- Our concept of an **AI-powered, chat-based skin advisor with real-time image input** was validated by both research trends and actual user feedback.

### 🔗 Answer to Our Research Objective

Our primary research question was:

*“Can AI improve the skincare shopping experience for new and unsure users?”*

Based on our findings — **yes**. AI can be an effective solution when:

- It’s designed with empathy for real user challenges,
- It provides clear, actionable guidance (not just product pushing), and
- It respects privacy and is easy to use.

### 🚀 Recommendations & Future Work

#### 1. Develop a working prototype

- Use simple UI tools like React and Firebase to create a test version of the chatbot.

#### 2. Test real-time camera input

- Train or integrate basic models to detect skin features (oily/dry/acne-prone).

### 3. Collaborate with domain experts

- Partner with dermatology students or professionals for accurate skincare logic.

### 4. Expand user testing

- Survey a larger and more diverse group for deeper insights.

### 5. Explore social integrations

- Many users follow influencers; future versions could include a way to compare AI advice with trending tips.

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### (APA Style)

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